

- $$\begin{array}{ll} \text{i)} & -1 \\ \text{ii)} & \frac{1}{24} \\ \text{iii)} & D \\ \text{iv)} & -1 \end{array}$$

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{x} - \frac{1}{x^2}}{\frac{1}{x}} = \frac{x-1}{x} = 1$$

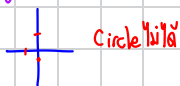
$$\begin{aligned} 2x \frac{dy}{dx} + 2y + 2y \frac{dy}{dx} &= 1 + \frac{y}{x} \\ (2x + 2y) \frac{dy}{dx} &= 1 - 2y \\ \frac{dy}{dx} &= \frac{1 - 2y}{2x + 2y - 1} = -\frac{1}{2} \end{aligned}$$

$$\frac{2}{x^3-x^2+x-1} = \frac{1}{x-1} + \frac{Bx+C}{x^2+1}$$

$$2 = A(x^2+1) + (Bx+C)(x-1)$$

$$\begin{aligned} x=1: \quad 2 &= 2A \rightarrow A=1 \rightarrow B=-1 \\ x=0: \quad 2 &= 1+C \rightarrow C=1 \end{aligned}$$

$$\begin{aligned} \frac{dy}{dx} &= y^2 + y + 1 \\ y &= \int_{-\infty}^{\infty} \frac{1}{1+t^2} dt \rightarrow \frac{dy}{dx} = -\frac{1}{1+t^2 x^2} \cdot \sec^2 x \\ (f^{-1})'(15) &= \frac{1}{f'(4)} = \frac{1}{48-24} = \frac{1}{24} \\ y' &= -\frac{x}{y} \end{aligned}$$



v) $L(x) = 2x \quad \sin(2 \cdot 0) + 2 \cos(2 \cdot 0)(x - 0)$

$$vi) \quad \frac{1}{x-1} + \frac{-x-1}{x^2+1}$$

$$3. i) \lim_{x \rightarrow 0} \frac{\sqrt{4+x} - \sqrt{8+x}}{x} = \lim_{x \rightarrow 0} \frac{\frac{1}{2\sqrt{4+x}} - \frac{1}{2\sqrt{8+x}}}{1} = \frac{\frac{1}{4} - \frac{1}{12}}{1} = \frac{1}{6}$$

$$ii) \lim_{x \rightarrow \infty} \frac{e^{2x} \arctan(e^x)}{e^{2x} + x} = \lim_{x \rightarrow \infty} \frac{\arctan e^x}{1 + \frac{x}{e^{2x}}} = \frac{\frac{\pi}{2}}{1} = \frac{\pi}{2}$$

$$\text{ii) } \lim_{n \rightarrow \infty} \sum_{k=1}^n \ln\left(\sqrt[n]{1+\frac{k}{n}}\right) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n} \ln\left(1+\frac{k}{n}\right) = \int_1^2 \ln x \, dx = [x \ln x - x]_1^2 = 2 \ln 2 - 2 + 1 = 2 \ln 2 - 1$$

$$4. \text{ i) } \int (x+10)(x-10)^{1/3} dx = \int (u+20) \cdot u^{1/3} du = \int (u^{4/3} + 20u^{1/3}) du = \frac{3}{7} u^{7/3} + 15 u^{4/3} + C = \frac{3}{7} (x-10)^{7/3} + 15 (x-10)^{4/3} + C$$

$$\text{ii) } \int \frac{1}{9x^2 + 6x + 5} dx = \int \frac{1}{(3x+1)^2 + 2^2} dx = \int \frac{1}{4(u^2 + 2^2)} \left(\frac{2}{3} du\right) = \frac{8}{3} \arctan u + C = \frac{8}{3} \arctan \left(\frac{3x+1}{2}\right) + C$$

$$\text{iii) } \int \frac{\ln x}{x} dx = -\frac{\ln x}{x} - \int -\frac{1}{x^2} dx = -\frac{\ln x}{x} - \frac{1}{x} + C$$

$$\text{iv) } \int \cos(\ln x) dx = x \cos(\ln x) + \int \sin(\ln x) dx = x \cos(\ln x) + x \sin(\ln x) - \int \cos(\ln x) dx$$

$$\therefore \int \cos(\ln x) dx = \frac{1}{2}(x \cos(\ln x) + x \sin(\ln x)) + C$$

$$v) \int \tan^3 \theta \sec^7 \theta d\theta = \int \tan^2 \theta \sec^6 \theta (\sec \theta \tan \theta d\theta) = \int (u^2 - 1)^3 \cdot u^6 du = \int u^8 - 2u^6 + u^4 du = \frac{\sec^9 \theta}{9} - \frac{2\sec^7 \theta}{7} + \frac{\sec^5 \theta}{5} + C$$

$$vi) \int_{2/\sqrt{3}}^3 \frac{1}{x^2 \sqrt{x^2-1}} dx = \frac{\cos^{-1}(1/3)}{\pi/3} \int_{\pi/3}^{\cos^{-1}(1/3)} \frac{1}{\sec^2 \theta \tan \theta} \cdot \sec^2 \theta \tan \theta d\theta = \frac{\cos^{-1}(1/3)}{\pi/3} \cos \theta d\theta = \left[\sin \theta \right]_{\pi/6}^{\cos^{-1}(1/3)} = \frac{2\sqrt{2}}{3} - \frac{\sqrt{3}}{2}$$

$$\begin{aligned} \text{vii)} \int_0^1 \frac{x^4(1-x^4)}{1+x^2} dx &= \int_0^1 \frac{x^6 - 4x^2 + 6x^0 - 4x^2 + x^4}{1+x^2} dx \\ &= \int_0^1 (x^6 - 4x^4 + 5x^2 - 4x^2 + 4 - \frac{4}{x^2}) dx \\ &= \left[\frac{x^7}{7} - \frac{4x^5}{5} + \frac{5x^3}{3} - \frac{4x^2}{2} + 4x - 4\arctan x \right]_0^1 \\ &= \frac{3-14+21-8+28}{21} - 4 \cdot \frac{\pi}{4} = \frac{66}{21} - \pi \end{aligned}$$

5. Curve length = $\int_4^{12} \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$

$$\frac{dx}{dy} = \frac{y}{8} - \frac{2}{y} = \int_4^{12} \left(\frac{y}{8} + \frac{2}{y} \right) dy$$

$$\frac{y^2}{64} + \frac{1}{2} + \frac{4}{y^2} = \left[\frac{y^2}{16} + 2 \ln y \right]_4^{12} = 8 + \ln 9$$

[illegible]

$$6. V = \int_{-\sqrt{3}/3}^{\sqrt{3}/3} \int \pi \cdot \left(\frac{1}{\sqrt{1+x^2}} \right) dx = \pi [\arctan x]_{-\sqrt{3}/3}^{\sqrt{3}/3} \\ = \pi \left(\frac{\pi}{3} - \left(-\frac{\pi}{6} \right) \right) = \frac{\pi^2}{2}$$

$$7. \text{ Let } y = \int_0^x \frac{1}{\sqrt{1+t^2}} dt = \int_0^{\tan^{-1}x} \frac{1}{\sec \theta} \cdot \sec^2 \theta d\theta = \left[\ln |\sec \theta + \tan \theta| \right]_0^{\tan^{-1}x} = \ln |\sqrt{1+x^2} + x| \\ \therefore \frac{dy}{dx} = \frac{1}{\sqrt{1+x^2}} \rightarrow \frac{dx}{dy} = \sqrt{1+x^2} \\ \frac{dx}{dy} = \frac{1}{2\sqrt{1+x^2}} (2x \frac{dx}{dy}) \\ = \frac{1}{2\sqrt{1+x^2}} \cdot 2x \sqrt{1+x^2} \\ = x \quad \therefore K=1$$

$$8. \text{ RHS: } \int_a^\infty x e^{-x} dx = [-x e^{-x} - e^{-x}]_a^\infty = (a+1)e^{-a} = e^{-a \ln(a+1)}$$

$$\text{LHS: } \lim_{x \rightarrow 0} (\cos x)^{2a/x} = \lim_{x \rightarrow 0} e^{\frac{2a}{x^2} \cdot \ln(\cos x)} = e^{\lim_{x \rightarrow 0} \frac{2a \ln(\cos x)}{x^2}} \\ \therefore \lim_{x \rightarrow 0} \frac{2a \ln(\cos x)}{x^2} = -a \ln(a+1) \quad (a \neq 0) \\ \lim_{x \rightarrow 0} \frac{2a \ln(\cos x)}{x^2} = -\ln(a+1) \quad (a \neq 0) \\ \lim_{x \rightarrow 0} \frac{2a \tan x}{2x} = \ln(a+1) \\ \lim_{x \rightarrow 0} \frac{\sec^2 x}{1} = \ln(a+1) \\ 1 = \ln(a+1) \rightarrow a = e-1$$

$$9. \frac{dy}{dx} = (2-x)(y^2+1) ; y(0) = 1$$

$$\int \frac{1}{y^2+1} dy = \int (2-x) dx$$

$$\arctan y = 2x - \frac{x^2}{2} + C$$

$$y = \tan(2x - \frac{x^2}{2} + C)$$

$$\therefore y(0)=1: 1 = \tan(0-0+C) \rightarrow C = \frac{\pi}{4}$$

$$\therefore y = \tan(2x - \frac{x^2}{2} + \frac{\pi}{4})$$

$$10. y' - \frac{\sin x}{1+\cos x} y = \frac{x^2}{1+\cos x} ; x \neq \pi$$

$$\therefore V(x) = e^{\int -\tan \frac{x}{2} dx} = e^{2 \ln(\cos \frac{x}{2})} = \cos^2 \frac{x}{2}$$

$$(\cos^2 \frac{x}{2}) y' - (\sin \frac{x}{2} \cos \frac{x}{2}) y = \frac{x^2}{2}$$

$$(\cos^2 \frac{x}{2} \cdot y)' = \frac{x^2}{2}$$

$$y \cdot \cos^2 \frac{x}{2} = \frac{x^3}{6} + C$$

$$y = \frac{x^3}{6} \sec^2 \frac{x}{2} + C \cdot \sec^2 \frac{x}{2}$$

$$11. i) s = \int_0^3 |2t-t^2| dt, t > 0 = \int_0^2 2t-t^2 dt + \int_2^3 t^2-2t dt \\ = [t^2 - \frac{t^3}{3}]_0^2 + [\frac{t^3}{3} - t^2]_2^3 = \frac{4}{3} + (0 + \frac{4}{3}) = \frac{8}{3}$$

$$ii) s(3) = 3^2 - \frac{3^3}{3} = 0 \rightarrow x\text{-coordinate is } 0 \text{ at } t=3 \text{ sec}$$

$$W = \int F \cdot ds = \int_0^3 -2(2t-t^2)(2t-t^2) dt$$

$$= -2 \int_0^3 4t^2 - 4t^3 + t^4 dt = -2 [\frac{4}{3}t^3 - t^4 + \frac{t^5}{5}]_0^3 = -2 [\frac{18}{5}]$$

$$\rightarrow \text{Work done to overcome the force} = 18 \text{ J}$$

12. $x = \tan y$

i) $V = \int_0^{\pi/4} \pi (\tan y)^2 dy$
 $= \pi \int_0^{\pi/4} \tan^2 y dy$
 $= \pi [\tan y - y]_0^{\pi/4} = \pi (1 - \frac{\pi}{4})$
 $\therefore T = \frac{\pi(1 - \frac{\pi}{4})}{\pi}$

ii) At t , $r(t) = V(t) = \int_0^h \pi \tan^2 y dy = \pi (\tan h - h)$

$\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$

$r = \pi \tan^2 h \cdot \frac{dh}{dt} \rightarrow \frac{dh}{dt} = \frac{r}{\pi \tan^2 h}$

At $h = \frac{\pi}{4}$ ($t = T$), $\frac{dh}{dt} = \frac{r}{\pi \cdot 1^2} = \frac{r}{\pi}$

13. i) $\therefore \lim_{x \rightarrow \infty} \frac{3^x}{1+3^{2x}} = \lim_{x \rightarrow \infty} \frac{3^{2x}}{1+3^{2x}} = 1$, $\int_2^{\infty} \frac{3^x}{1+3^{2x}} dx$ and $\int_0^{\infty} \frac{1}{3^x}$ both diverge or converge in the same way.

Converges

$\left[-\frac{1}{3^x \ln 3} \right]_0^{\infty} = \frac{1}{\ln 3}$ Converges

ii) $\lim_{x \rightarrow 0} \frac{x^2}{\ln^2(x+1)} = \frac{2x}{\frac{2\ln(x+1)}{x+1}} = \frac{2}{\frac{2\ln(x+1)}{(x+1)^2}} = \frac{2}{\frac{2-2\ln(x+1)}{(x+1)^2}} = \frac{2}{2} = 1$

$\therefore \lim_{x \rightarrow 0} \frac{\ln^2(x+1)^3 \sqrt{x^2 + \sin x}}{x^{4/3}} = \lim_{x \rightarrow 0} \frac{\ln^2(x+1)}{x^2} \cdot \frac{\sqrt{x^2 + \sin x}}{x^{2/3}} = \lim_{x \rightarrow 0} \frac{\ln^2(x+1)}{x^2} \cdot \lim_{x \rightarrow 0} \frac{x^{2/3} \sqrt{\frac{\sin x}{x}}}{x^{2/3}}$
 $= \lim_{x \rightarrow 0} \frac{2\ln(x+1)}{x+1} = \lim_{x \rightarrow 0} \frac{2}{x+1} = \frac{2}{1} = 2$
 $= \lim_{x \rightarrow 0} \frac{1 - \ln(0+1)}{1^2} = 1$

$\therefore \frac{\ln^2(x+1)^3 \sqrt{x^2 + \sin x}}{x^{4/3}}$ and $\frac{1}{x^{4/3}}$ grows at the same rate

$\int_0^1 \frac{\ln^2(x+1)^3 \sqrt{x^2 + \sin x}}{x^{4/3}} dx$ & $\int_0^1 \frac{1}{x^{4/3}} dx$ both conv/div at the same time
 $= [-3x^{-1/3}]_0^1 = \infty$ Diverges

Diverges

14. $\int_0^4 x e^{(x-2)^4} dx = \int_{-2}^2 (u+2) e^{u^4} du = \int_{-2}^2 u e^{u^4} du + \underbrace{2 \int_{-2}^2 e^{u^4} du}_K = 2K$

$u = x-2 \rightarrow du = dx$

$\int_{-2}^2 e^{u^4} du = K$

$\therefore f(u) = u e^{u^4}$ is an odd function, $\int_{-2}^2 u e^{u^4} du = 0$
 $f(-u) = -u e^{(-u)^4} = -u e^{u^4} = -f(u)$